WP3 Solutions

247. $P(1,-2,3)$, $v = \langle 1,2,3 \rangle$

A) $x = 1+t,\ y = -2+2t,\ z = 3+3t$

B) $x-1 = \frac{y+2}{2} = \frac{z-3}{3}$

C) if this intersects the xy-plane then $z=0$ which means that $t=-1$ so we get $(0,-4,0)$

257. $P(3,1,0)$ $Q(1,4,-3)$ want $\perp$ to $x=3t,\ y=3+8t,\ z=-7+6t$

$v = \langle 1-3, 4-1, -3-0 \rangle$
$= \langle -2,3,-3 \rangle$

let v_1 be the vector from the line

$v_1 = \langle 3,8,6 \rangle$

so $v \cdot v_1 = -2(3)+3(8)-3(6)$
$= -6+24-18$
$= 0$

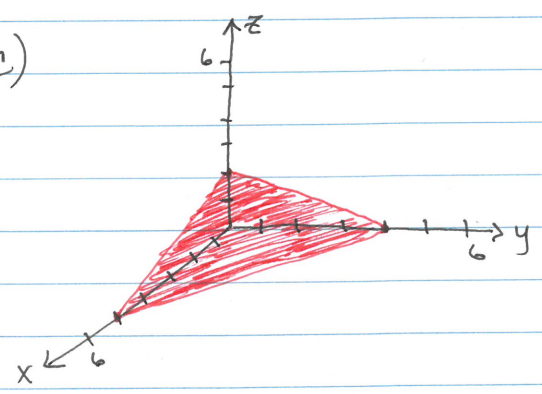
So these are perpendicular.

271. $4x+5y+10z-20=0$

A) $n = 4\hat{i}+5\hat{j}+10\hat{k}$

B) $x,y=0$: $10z=20$ so $z=2$; $(0,0,2)$
$y,z=0$: $4x=20$ so $x=5$; $(5,0,0)$
$x,z=0$: $5y=20$ so $y=4$; $(0,4,0)$

C)



251. $x=1+t,\ y=3+t,\ z=5+4t$

A) $P(1,3,5)$ and $v = \langle 1,1,4 \rangle$

B) $\vec{OP} = \langle 0-1, 0-3, 0-5 \rangle$
$= \langle -1,-3,-5 \rangle$

$$\vec{OP} \times v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & -3 & -5 \\ 1 & 1 & 4 \end{vmatrix}$$
$= (-12+5)\hat{i} - (-4+5)\hat{j} + (-1+3)\hat{k}$
$= -7\hat{i} - \hat{j} + 2\hat{k}$

$$d = \frac{\sqrt{(-7)^2+(-1)^2+2^2}}{\sqrt{1^2+1^2+4^2}}$$
$= \sqrt{54}/\sqrt{18}$
$= \sqrt{3}$

269. $P(1,2,3)$ $n = \langle 1,2,3 \rangle$

A) $1(x-1)+2(y-2)+3(z-3)=0$

B) $x-1+2y-4+3z-9=0$
$x+2y+3z-14=0$

279. Show $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-2}{4}$ is parallel to $x-2y+z=6$

$x=2t+1$, $y=3t-1$, $z=4t+2$

$2t+1-2(3t-1)+(4t+2)=6$

$2t+1-6t+2+4t+2=6$

$5=6$ False

So there are no points of intersection between the two lines, hence they are parallel.

285. $P(-1,2,1)$ perpendicular intersection of $x+y+z-2=0$ and

$2x-y+3z-1=0$

$n_1 = \langle 1,1,-1 \rangle$ and $n_2 = \langle 2,-1,3 \rangle$

so the normal vector of the intersection is

$n = n_1 \times n_2$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 2 & -1 & 3 \end{vmatrix}$$

$= (3-1)\hat{i} - (3+2)\hat{j} + (-1-2)\hat{k}$

$n = \langle 2,-5,-3 \rangle$

So the plane through P is

$2(x+1)-5(y-2)-3(z-1)=0$

$2x+2-5y+10-3z+3=0$

$2x-5y-3z+15=0$